Wen-Chi (Vicky) Tsai

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Work Experience

Research Assistant, Agent Harness and Model Co-Optimization — Sep. 2026 – Present
Language Technologies Institute, CMU, advised by Prof. Graham Neubig — *Pittsburgh, PA*

- Owned the **evaluation and data-selection track** of a research project that automatically improves **LLM agents**, building the Python tooling that tests each agent version and decides which problems it is measured on
- Found the project's real bottleneck was **evaluation reliability**: the benchmark disagreed with itself on **42% of problems** under identical re-runs (Cohen's kappa 0.16), explaining three null experiments
- Ran the lab's **self-improving agent loop** end to end for the first time on the GPU cluster, fixing a breakage on main in a **merged pull request** so the system proposed its first new agent version from baseline failures

Software Engineering Intern — Jun. 2025 – Mar. 2026
ASUS, Open Cloud Infrastructure Software R&D Team — *Taipei, Taiwan*

- Automated issue triage with a **GitLab Model Context Protocol (MCP) agent**, pairing an LLM classification prompt with MCP tool calls to auto-classify, label, and route issues, then backfilling the manual backlog
- Shipped four **context providers into a VS Code AI coding assistant**, adding commit-history, OS-metadata, current-file, and repository-map context selected from an evaluation of candidates
- Cut a full regression pass from **∼16 hours to ∼1–2 hours** with a **TypeScript end-to-end (E2E) test suite** covering ∼95% of manual cases across Linux, Windows, and macOS

Research Assistant, Edge LLM Compression — Sep. 2025 – Feb. 2026
Wireless Ad-Hoc and Sensor Networks Laboratory, NCU — *Taoyuan, Taiwan*

- Recovered **4-bit accuracy at 2-bit model size** (IoU 0.9998, 1.18 GB) by fine-tuning a Llama-3.2 student with **GPTQ quantization and LLM-Neo knowledge distillation** through a LoRA adapter in PyTorch
- Evaluated each configuration against **4-bit and 2-bit quantized baselines**, tuning **LoRA rank and KD loss weighting** to isolate how much accuracy distillation recovered

Startup Engineer — Jun. 2026 – Aug. 2026
RoomMuse — *Pittsburgh, PA*

- Caught unbuildable layouts at design time rather than on install day with a **six-rule geometric conflict engine** that blocks any unit sealing a socket, pipe, or doorway while passing intended connections (**45 tests**)

Selected Projects

Museum Artifact Recognition App — May 2024 – Nov. 2024

- Shipped **on-device model inference in an iOS app**, running the team's YOLOv8n and Pix2Pix reflection-removal models in **Swift with TensorFlow Lite** at 94.5% accuracy across 7 classes
- Curated the **2,000-image training set and 699-image evaluation set** spanning varied angles and lighting, and added reposition guidance so the app recovers from failed recognitions

Dual-Strategy UAV Path Planning — Sep. 2024 – Dec. 2025

- Designed a dual-strategy planner trading **layered MST-DFS** for structured-space traversal against **collision-aware A* replanning**, cutting path-generation time over 5× and path length up to 14%
- Built the **3D point-cloud pipeline in C++ and Python**, slicing raw .pcd scans into layers and chaining **per-layer KNN and MST** into a DFS traversal, with normal-vector correction for full coverage

Publications

Wen-Chi Tsai, Yi-Peng Liu, Yu Wu, Yun-Sheng Chen, Chung-Lun Lai, Yu-Yu Chang, Min-Te Sun. "UAV Path Planning for Building Structure Inspection." In *14th International Conference on Ubi-Media Computing (Ubi-Media 2026)*, Penang, Malaysia.

Education

Carnegie Mellon University (CMU), School of Computer Science — Aug. 2026 – Dec. 2027 (Expected)
Master of Science in Intelligent Information Systems, Language Technologies Institute — *Pittsburgh, PA*

- **Relevant coursework:** AI Agents, Multimodal Machine Learning, Intro to Question Answering with LLMs

National Central University (NCU) — Sep. 2022 – Jun. 2026
Bachelor of Science in Computer Science and Information Engineering | GPA: 3.99/4.30 — *Taoyuan, Taiwan*

- **Relevant coursework:** Generative AI (Text & Image), Pattern Recognition, Data Science, Software Engineering